

STUDIU ASUPRA INTENSITĂȚII SEMNALULUI ÎN ECOGRAFII ABDOMINALE

Burlacu Claudiu-Ionuț



Universitatea
Transilvania
din Brașov

FACULTATEA DE ȘTIINȚE ECONOMICE
ȘI ADMINISTRAREA AFACERILOR





Cuprins

- 1. Obiectivul studiului
- 2. Tehnologiile utilizate
- 3. Etapele de dezvoltare ale studiului
- 4. Concluzie





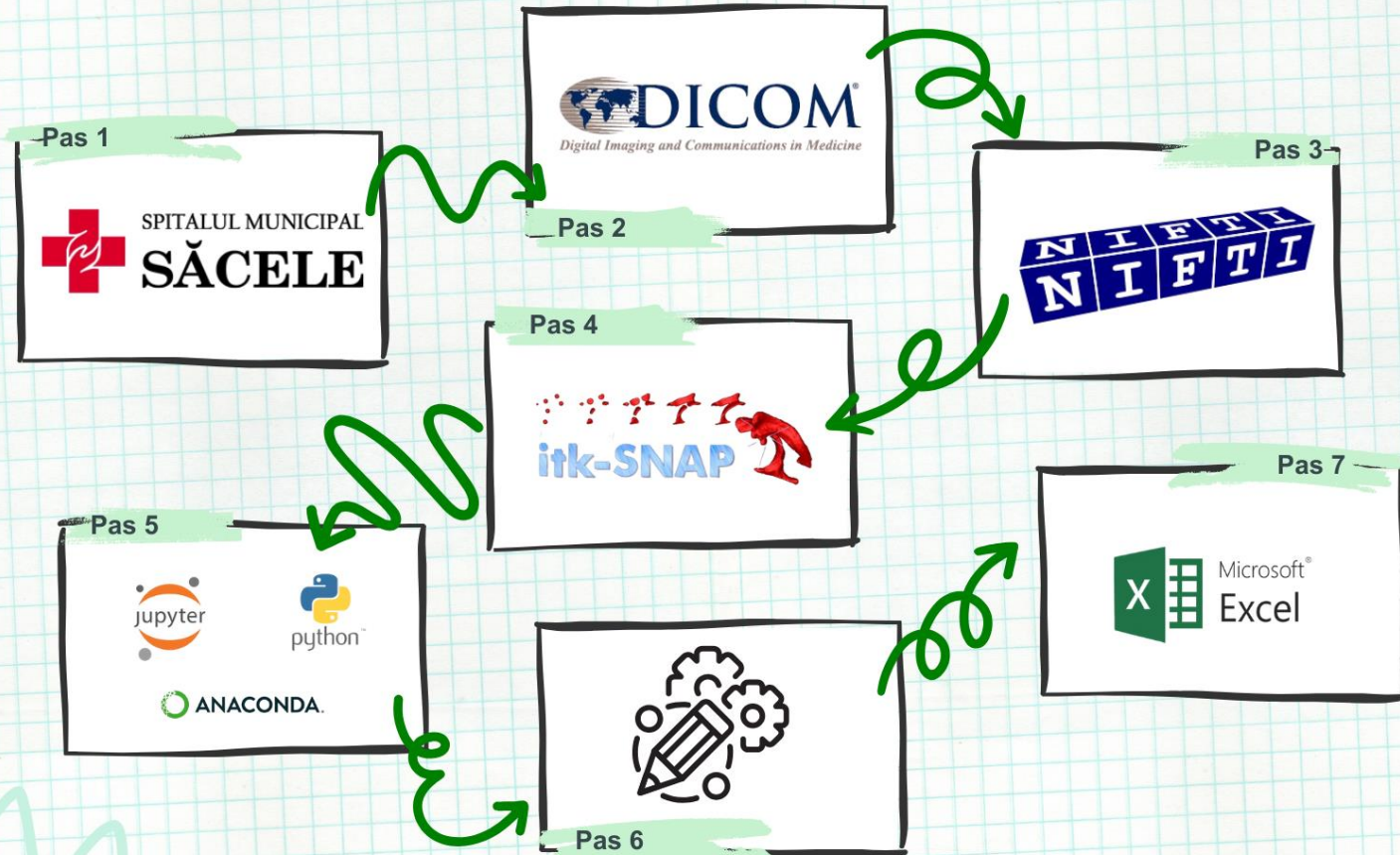
1. Obiectivul studiului

- Analiză cantitativă a imaginilor ecografice abdominale.
- Scopul studiului este analiza variației semnalului ecografic în funcție de adâncime, în zonele ficatului și rinichilor.
- Ipoteză: intensitatea semnalului ecografic scade odată cu creșterea adâncimii.





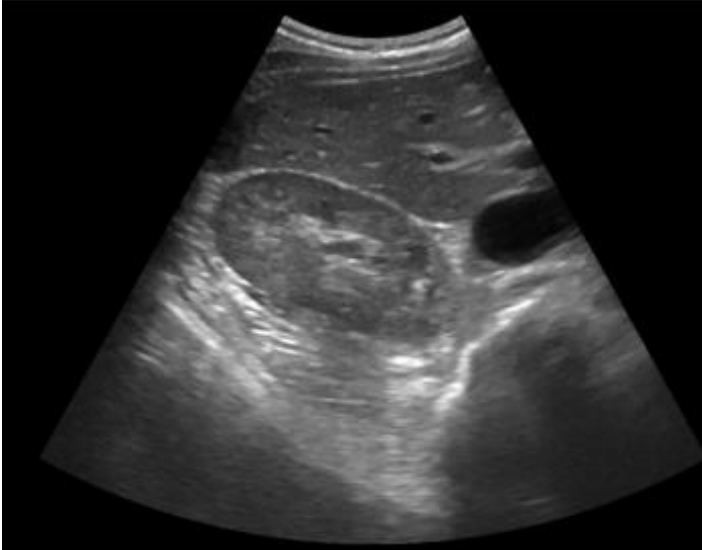
2. Tehnologiile utilizate



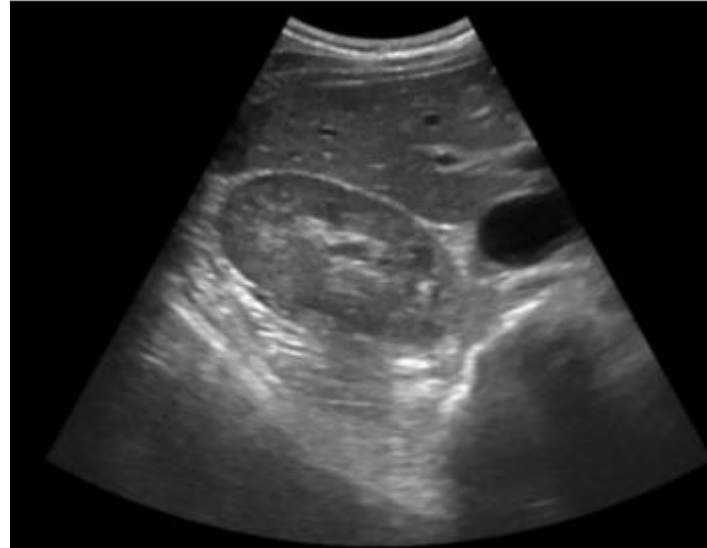


3. Etapele de dezvoltare ale studiului

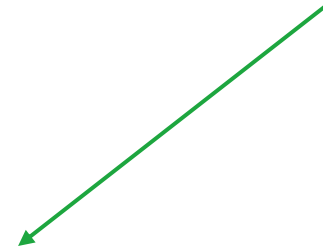
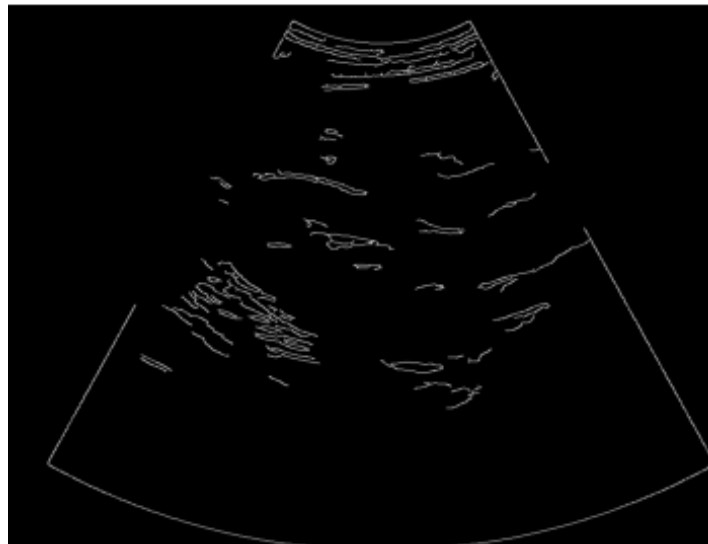
Original Image



Blurred Image



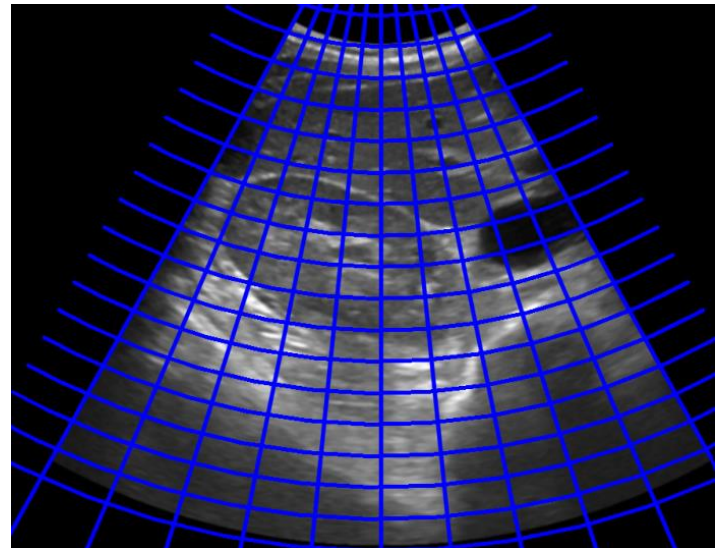
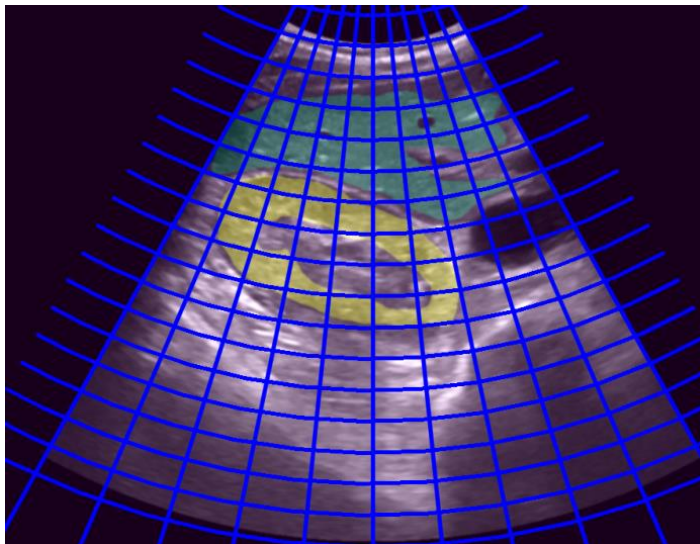
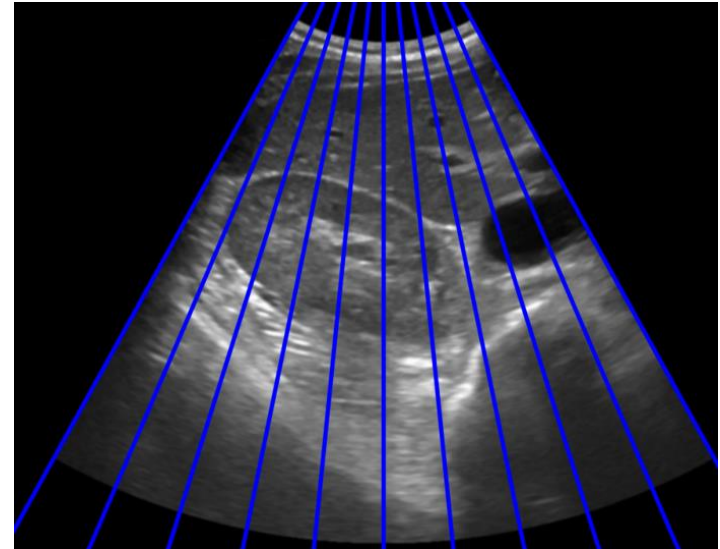
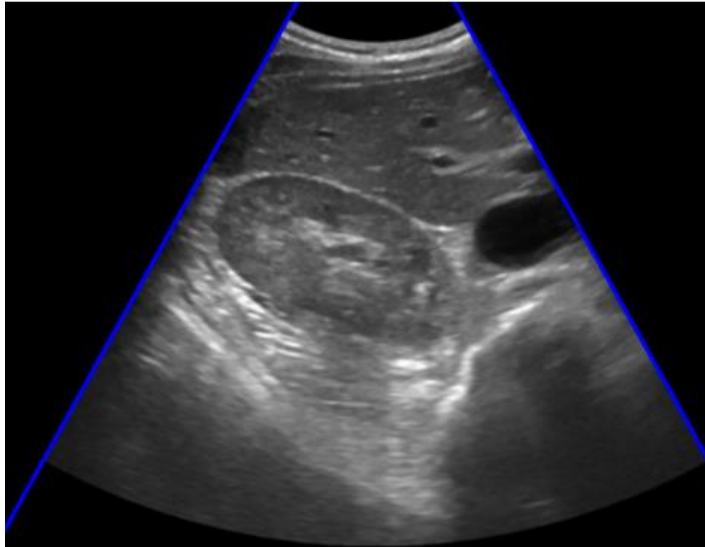
Canny Edge Detection





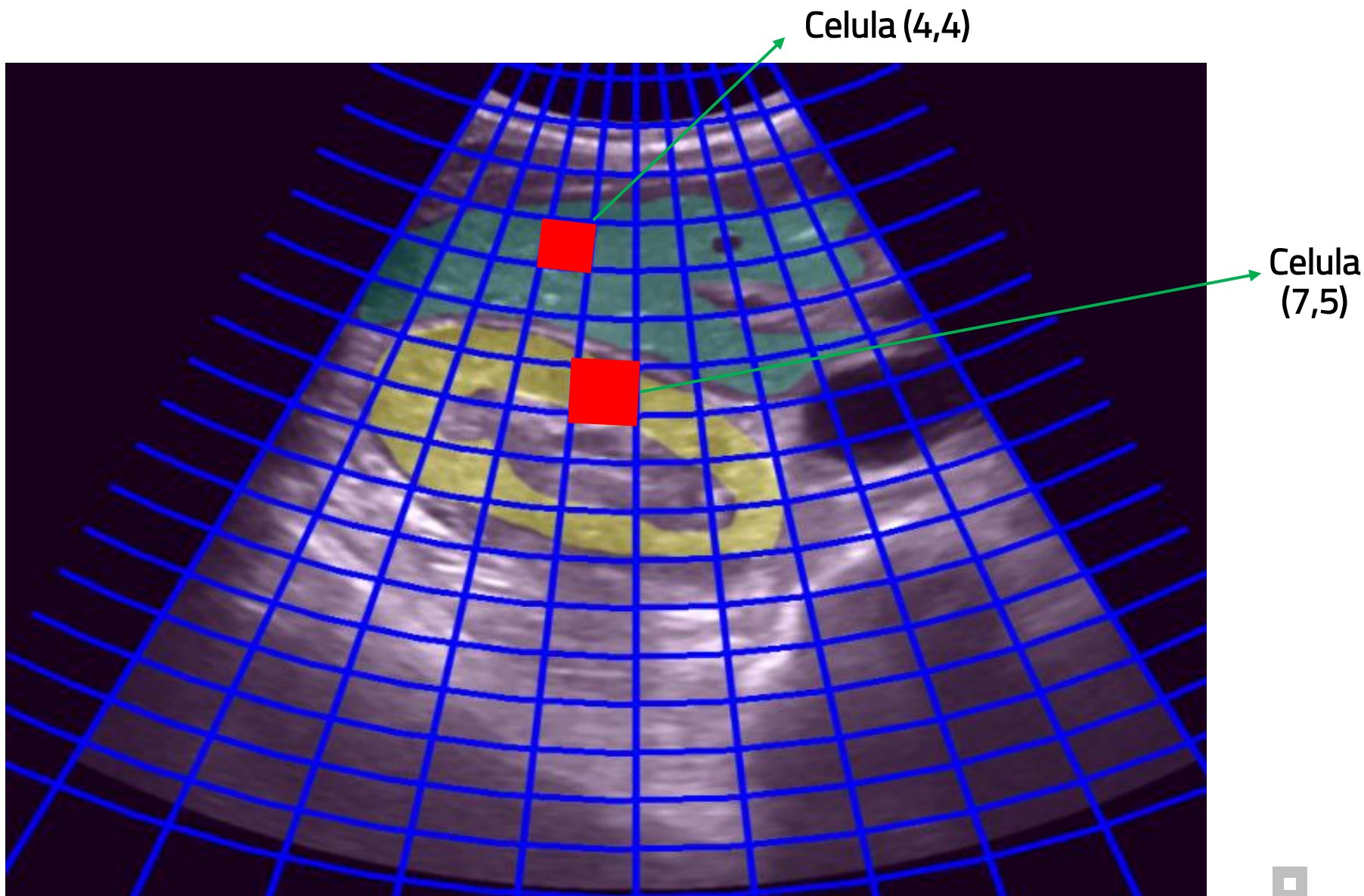
3. Etapele de dezvoltare ale studiului

Hough Transform



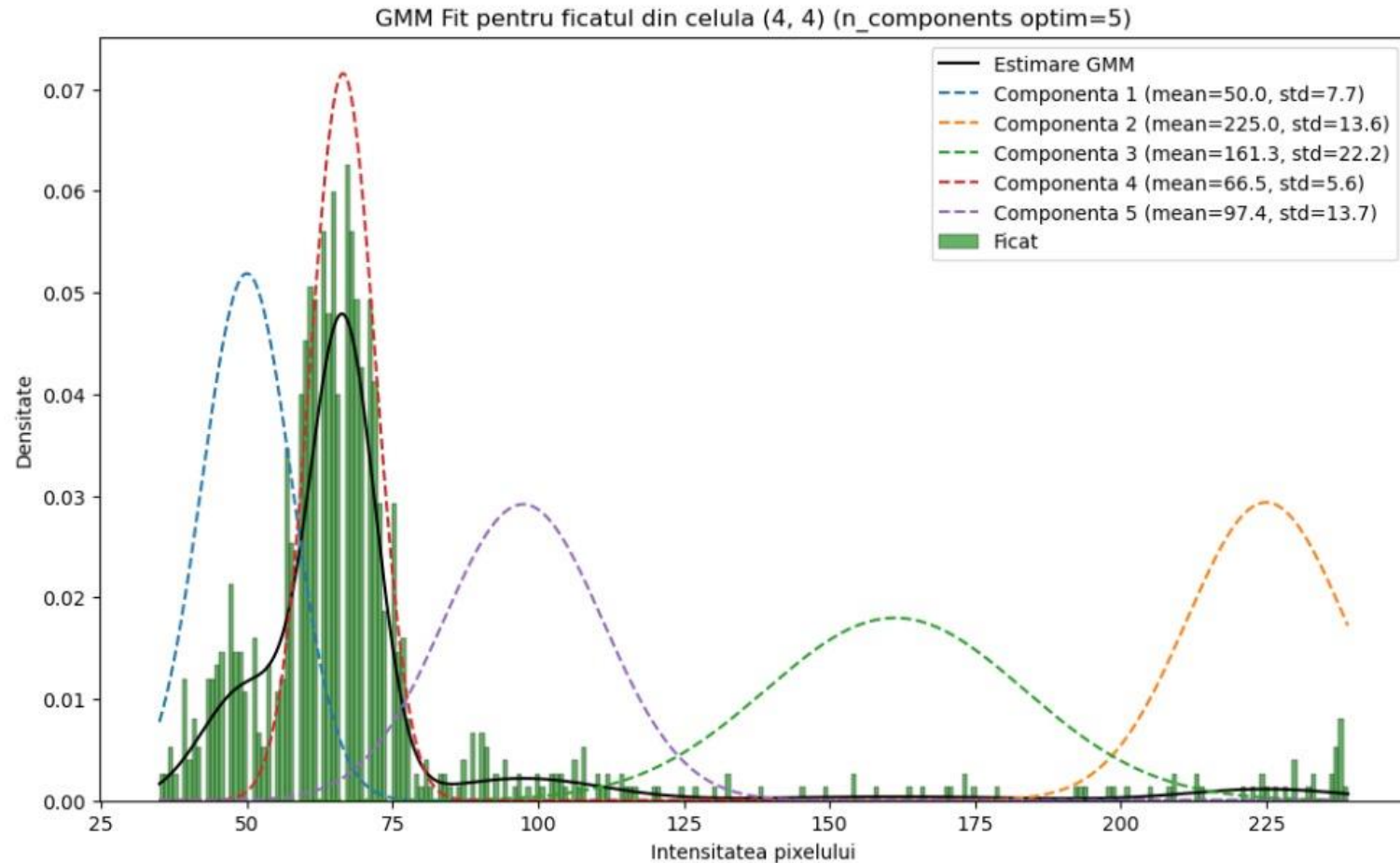


3. Etapele de dezvoltare ale studiului



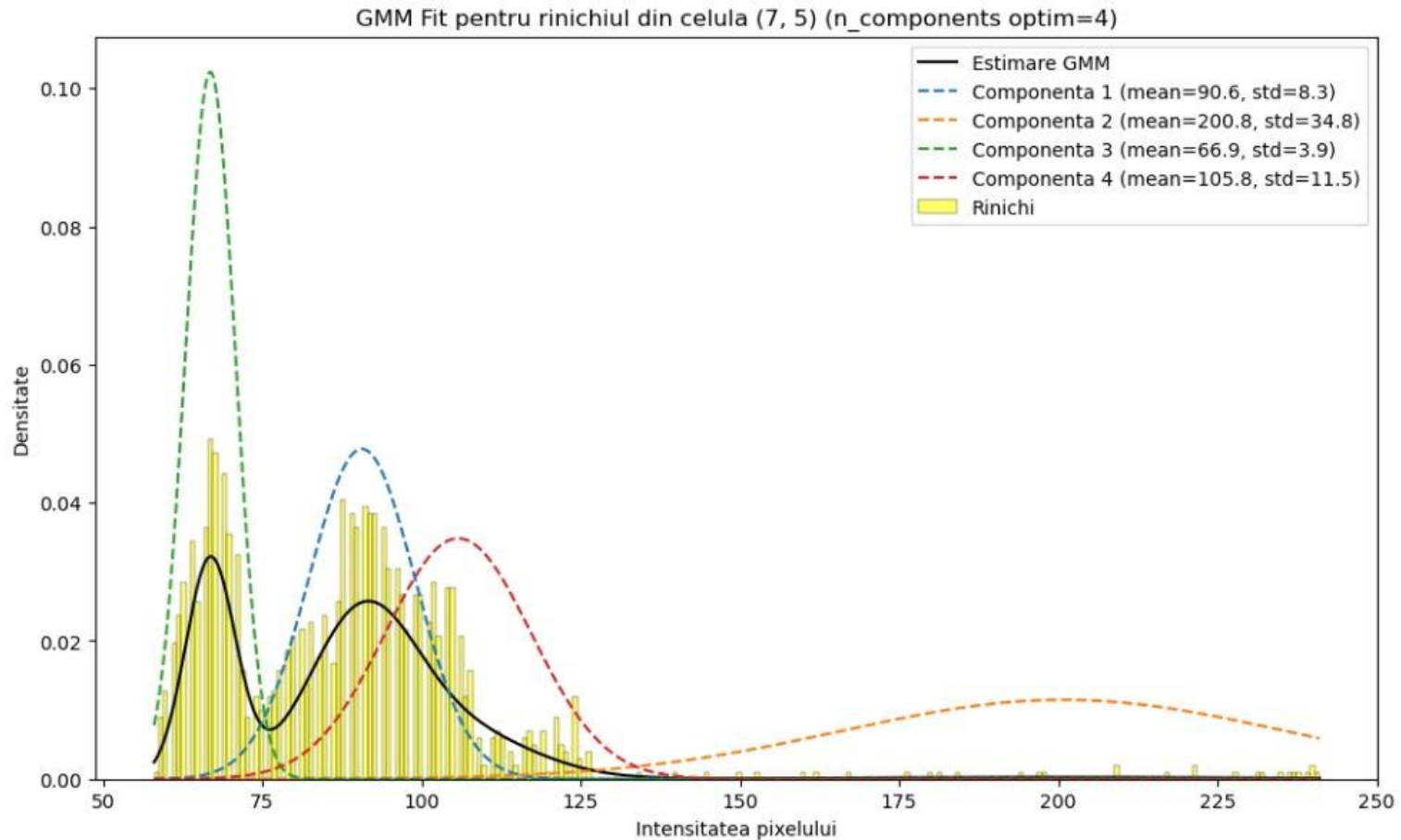


3. Etapele de dezvoltare ale studiului





3. Etapele de dezvoltare ale studiului





3. Etapele de dezvoltare ale studiului

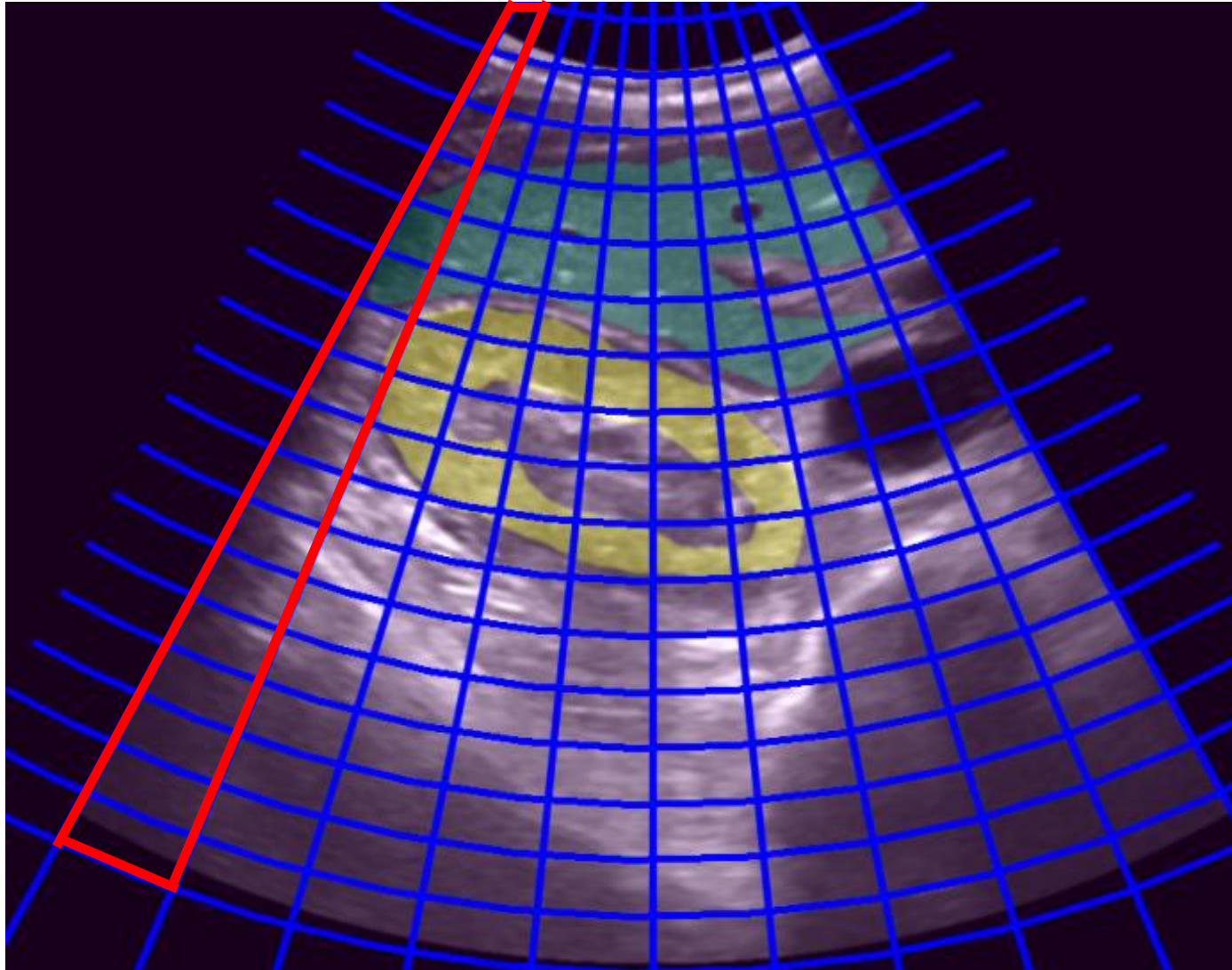
	A	B	C	D	E	F	G	H	I	J	K
1	Patient	Image	Organ	Cell_coord	Nr_moduri	Id_Mod	Mean	Std	Total_pixels	Liver_pixels	Kidney_pixels
2	AT082158	echography_AT082158_22_bw.png	Liver	(3, 1)	2	1	65.09274861	5.178206379	1263	619	0
3	AT082158	echography_AT082158_22_bw.png	Liver	(3, 1)	2	2	52.63727272	6.06653704	1263	619	0
4	AT082158	echography_AT082158_22_bw.png	Liver	(4, 1)	3	1	56.00397678	4.89024303	1430	1303	0
5	AT082158	echography_AT082158_22_bw.png	Liver	(4, 1)	3	2	42.79150438	6.220557495	1430	1303	0
6	AT082158	echography_AT082158_22_bw.png	Liver	(4, 1)	3	3	68.0206211	12.9934669	1430	1303	0
7	AT082158	echography_AT082158_22_bw.png	Liver	(5, 1)	3	1	29.260162	5.861415006	1597	1438	0
8	AT082158	echography_AT082158_22_bw.png	Liver	(5, 1)	3	2	40.23804317	4.7267997	1597	1438	0
9	AT082158	echography_AT082158_22_bw.png	Liver	(5, 1)	3	3	61.524916	13.17612079	1597	1438	0
10	AT082158	echography_AT082158_22_bw.png	Liver	(6, 1)	2	1	45.34866714	5.819125598	1728	1555	0
11	AT082158	echography_AT082158_22_bw.png	Liver	(6, 1)	2	2	31.59956502	6.665691311	1728	1555	0
12	AT082158	echography_AT082158_22_bw.png	Liver	(7, 1)	4	1	32.85574603	5.977409	1935	1728	0
13	AT082158	echography_AT082158_22_bw.png	Liver	(7, 1)	4	2	53.47417082	4.145201438	1935	1728	0
14	AT082158	echography_AT082158_22_bw.png	Liver	(7, 1)	4	3	79.70215822	10.82761145	1935	1728	0
15	AT082158	echography_AT082158_22_bw.png	Liver	(7, 1)	4	4	44.13115474	3.728092343	1935	1728	0
16	AT082158	echography_AT082158_22_bw.png	Liver	(8, 1)	3	1	54.07420304	5.315376513	2044	1390	0
17	AT082158	echography_AT082158_22_bw.png	Liver	(8, 1)	3	2	44.75701738	7.40034321	2044	1390	0
18	AT082158	echography_AT082158_22_bw.png	Liver	(8, 1)	3	3	76.08261696	7.880426643	2044	1390	0
19	AT082158	echography_AT082158_22_bw.png	Liver	(9, 1)	1	1	55.79166667	7.838671249	2255	1944	0
20	AT082158	echography_AT082158_22_bw.png	Liver	(10, 1)	3	1	58.96000063	4.197507373	2404	1870	0
21	AT082158	echography_AT082158_22_bw.png	Liver	(10, 1)	3	2	72.97794515	7.872018732	2404	1870	0
22	AT082158	echography_AT082158_22_bw.png	Liver	(10, 1)	3	3	48.52063091	4.680309076	2404	1870	0
23	AT082158	echography_AT082158_22_bw.png	Liver	(11, 1)	1	1	61.80798911	8.207803313	2527	2203	0
24	AT082158	echography_AT082158_22_bw.png	Liver	(12, 1)	1	1	60.3593947	7.496607609	2733	2379	0
25	AT082158	echography_AT082158_22_bw.png	Liver	(13, 1)	2	1	57.14011549	4.686259358	2844	810	0
26	AT082158	echography_AT082158_22_bw.png	Liver	(13, 1)	2	2	69.60613166	4.536182927	2844	810	0
27	AT082158	echography_AT082158_22_bw.png	Liver	(3, 2)	2	1	75.18604214	2.790596031	1263	483	0
28	AT082158	echography_AT082158_22_bw.png	Liver	(3, 2)	2	2	70.18322516	1.899423096	1263	483	0

Structura tabelului Excel rezultat din analiza GMM





3. Etapele de dezvoltare ale studiului





3. Etapele de dezvoltare ale studiului

	A	B	C	D	E	F	G	H
1	pacient	imagine	coloana	celula	organ	media_gri_celula_1	media_gri_celula_2	panta
2	AT082158	echography_AT082158_22_bw.png	1 3-4	Liver		59.96122779	53.63699156	-6.324236229
3	AT082158	echography_AT082158_22_bw.png	1 3-5	Liver		59.96122779	38.4123783	-10.77442474
4	AT082158	echography_AT082158_22_bw.png	1 3-6	Liver		59.96122779	39.53697749	-6.808083432
5	AT082158	echography_AT082158_22_bw.png	1 3-7	Liver		59.96122779	44.8587963	-3.775607873
6	AT082158	echography_AT082158_22_bw.png	1 3-8	Liver		59.96122779	52.64388489	-1.463468579
7	AT082158	echography_AT082158_22_bw.png	1 3-9	Liver		59.96122779	55.79166667	-0.694926853
8	AT082158	echography_AT082158_22_bw.png	1 3-10	Liver		59.96122779	58.47219251	-0.212719325
9	AT082158	echography_AT082158_22_bw.png	1 3-11	Liver		59.96122779	61.80798911	0.230845165
10	AT082158	echography_AT082158_22_bw.png	1 3-12	Liver		59.96122779	60.3593947	0.044240769
11	AT082158	echography_AT082158_22_bw.png	1 3-13	Liver		59.96122779	63.42469136	0.346346357
12	AT082158	echography_AT082158_22_bw.png	1 4-5	Liver		53.63699156	38.4123783	-15.22461325
13	AT082158	echography_AT082158_22_bw.png	1 4-6	Liver		53.63699156	39.53697749	-7.050007033
14	AT082158	echography_AT082158_22_bw.png	1 4-7	Liver		53.63699156	44.8587963	-2.926065087
15	AT082158	echography_AT082158_22_bw.png	1 4-8	Liver		53.63699156	52.64388489	-0.248276666
16	AT082158	echography_AT082158_22_bw.png	1 4-9	Liver		53.63699156	55.79166667	0.430935022
17	AT082158	echography_AT082158_22_bw.png	1 4-10	Liver		53.63699156	58.47219251	0.805866826
18	AT082158	echography_AT082158_22_bw.png	1 4-11	Liver		53.63699156	61.80798911	1.167285364
19	AT082158	echography_AT082158_22_bw.png	1 4-12	Liver		53.63699156	60.3593947	0.840300393
20	AT082158	echography_AT082158_22_bw.png	1 4-13	Liver		53.63699156	63.42469136	1.0875222
21	AT082158	echography_AT082158_22_bw.png	1 5-6	Liver		38.4123783	39.53697749	1.124599189
22	AT082158	echography_AT082158_22_bw.png	1 5-7	Liver		38.4123783	44.8587963	3.223208997
23	AT082158	echography_AT082158_22_bw.png	1 5-8	Liver		38.4123783	52.64388489	4.74383553
24	AT082158	echography_AT082158_22_bw.png	1 5-9	Liver		38.4123783	55.79166667	4.344822091
25	AT082158	echography_AT082158_22_bw.png	1 5-10	Liver		38.4123783	58.47219251	4.011962842
26	AT082158	echography_AT082158_22_bw.png	1 5-11	Liver		38.4123783	61.80798911	3.899268467
27	AT082158	echography_AT082158_22_bw.png	1 5-12	Liver		38.4123783	60.3593947	3.135288057
28	AT082158	echography_AT082158_22_bw.png	1 5-13	Liver		38.4123783	63.42469136	3.126539132

Structura tabelului Excel rezultat din analiza gradientului





3. Etapele de dezvoltare ale studiului

	A	B	C	D
1	pacient	organ	adancime	media
2	07-09-2023-0003_20240115s	Liver	2.5	105.440613
3	07-09-2023-0003_20240115s	Liver	3	105.5774878
4	07-09-2023-0003_20240115s	Liver	3.5	116.4321318
5	07-09-2023-0003_20240115s	Liver	4	119.5236511
6	07-09-2023-0003_20240115s	Liver	4.5	120.0955779
7	07-09-2023-0003_20240115s	Liver	5	118.2720217
8	07-09-2023-0003_20240115s	Liver	5.5	114.3463548
9	07-09-2023-0003_20240115s	Liver	6	109.1757661
10	07-09-2023-0003_20240115s	Liver	6.5	106.6742538
11	07-09-2023-0003_20240115s	Liver	7	102.3100478
12	07-09-2023-0003_20240115s	Liver	7.5	101.4893974
13	07-09-2023-0003_20240115s	Liver	8	99.81009967
14	07-09-2023-0003_20240115s	Liver	8.5	97.17273462
15	07-09-2023-0003_20240115s	Liver	9	95.56397972
16	07-09-2023-0003_20240115s	Liver	9.5	91.97389544
17	07-09-2023-0003_20240115s	Liver	10	90.72661299
18	07-09-2023-0003_20240115s	Liver	10.5	87.22424677
19	07-09-2023-0003_20240115s	Liver	11	85.88228392
20	07-09-2023-0003_20240115s	Liver	11.5	81.23830813
21	07-09-2023-0003_20240115s	Liver	12	77.32088933
22	07-09-2023-0003_20240115s	Liver	12.5	72.7146903
23	07-09-2023-0003_20240115s	Liver	13	70.37373465
24	07-09-2023-0003_20240115s	Liver	13.5	67.18639908
25	07-09-2023-0003_20240115s	Liver	14	66.53722279
26	07-09-2023-0003_20240115s	Liver	14.5	64.18540469
27	07-09-2023-0003_20240115s	Liver	15	64.17662338
28	07-09-2023-0003_20240115s	Liver	15.5	60.98400341

Structura tabelului Excel rezultat din analiza agregată

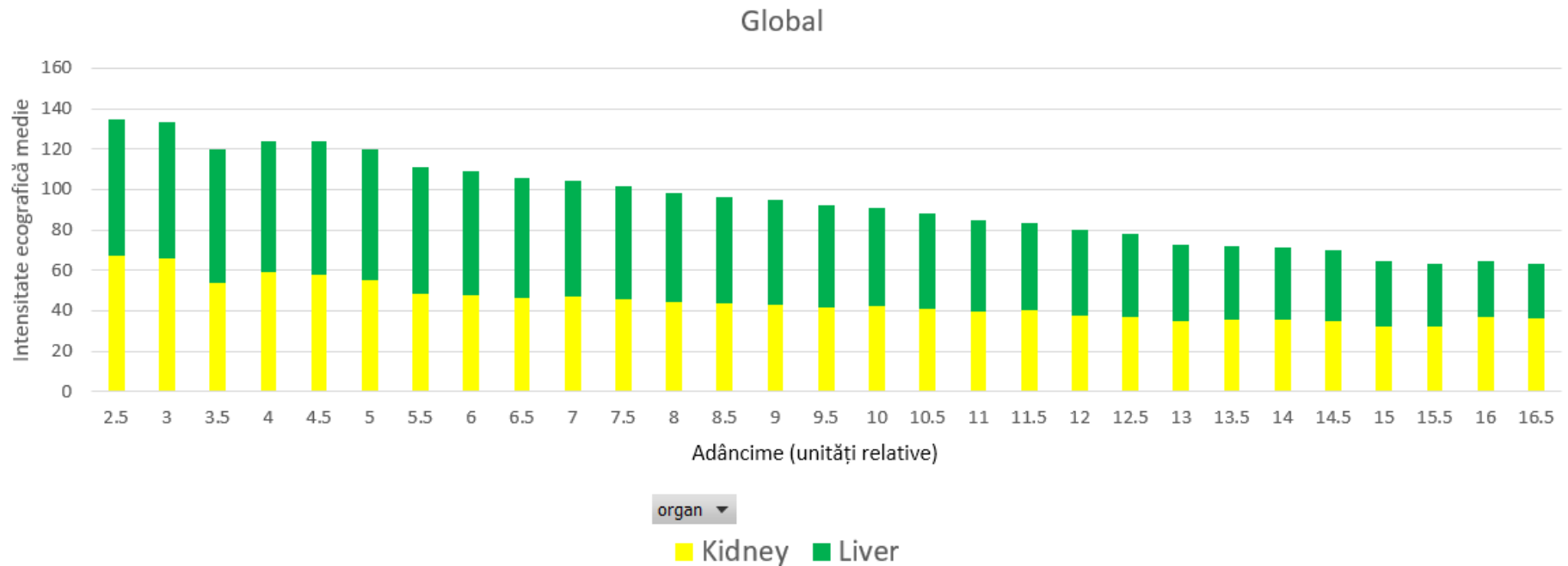




3. Etapele de dezvoltare ale studiului

pacient ▼

Average of media



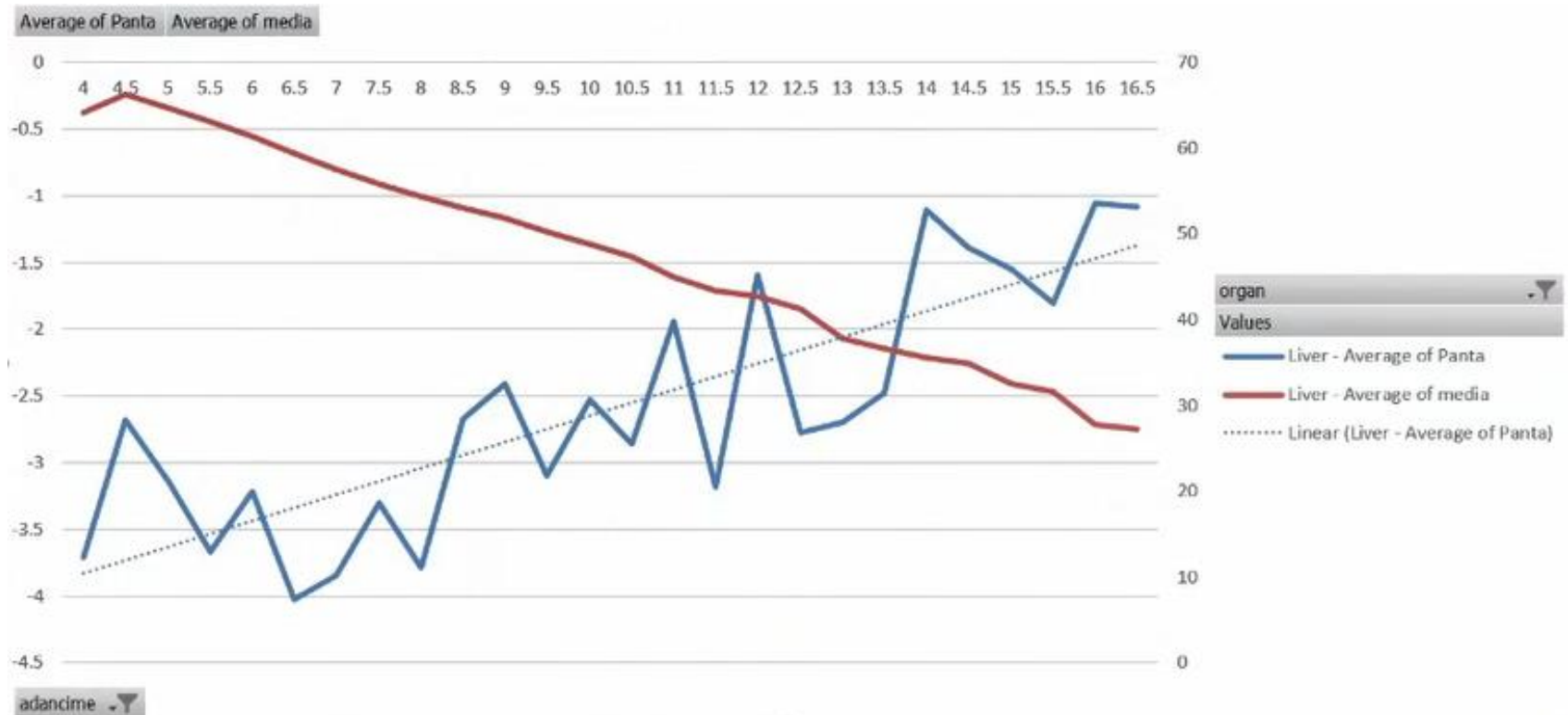
adancime ▼

Media intensității ecografice pe adâncime în urma analizei gradientului vertical





3. Etapele de dezvoltare ale studiului



Evoluţia mediei intensităţii ecografice şi a pantei gradientului vertical în funcţie de adâncime (ficat)



4. Concluzii

- ❑ Prototip metodă computațională.
- ❑ Ipoteza de lucru – confirmată: intensitatea semnalului scade în funcție de adâncime.
- ❑ Rezultatele obținute sunt cantitative, reproductibile și structurate.
- ❑ Metodă cu potențial pentru sisteme automate de evaluare imagistică.